

LightGBM (No EVI/NDVI)

Acer Pollen Prediction – Original Scale

(Duan Smearing Back-transformation)

Pooled model (all stations), 5-seed average, early stopping, Bayesian-optimized

Duan Smearing Factor: 1.0257

Test Performance (Original Scale):

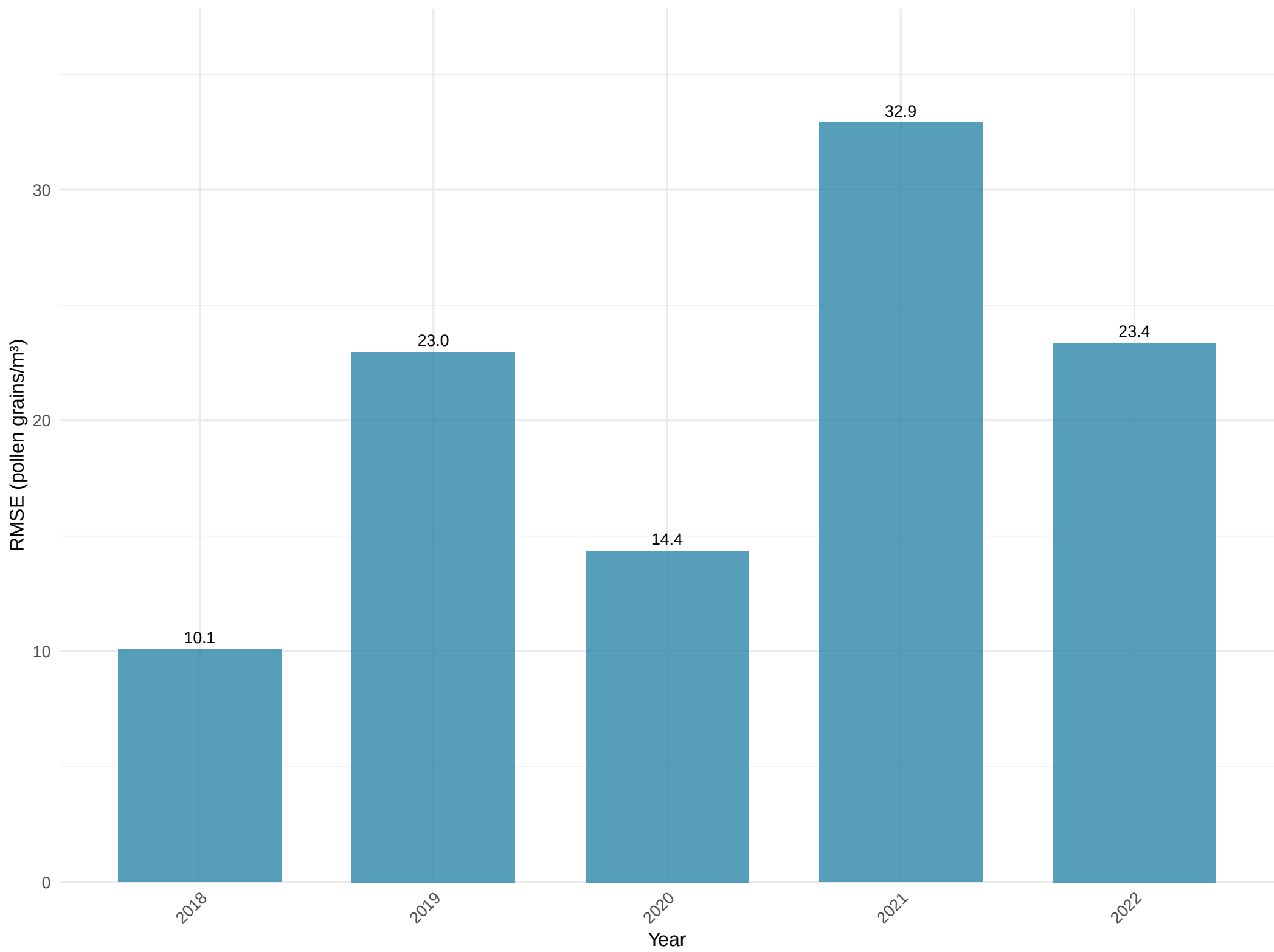
RMSE: 18.54 | MAE: 4.24 | R^2 : 0.7631

Actual vs Predicted (Original Scale)

RMSE: 18.54 | MAE: 4.24 | R²: 0.7631

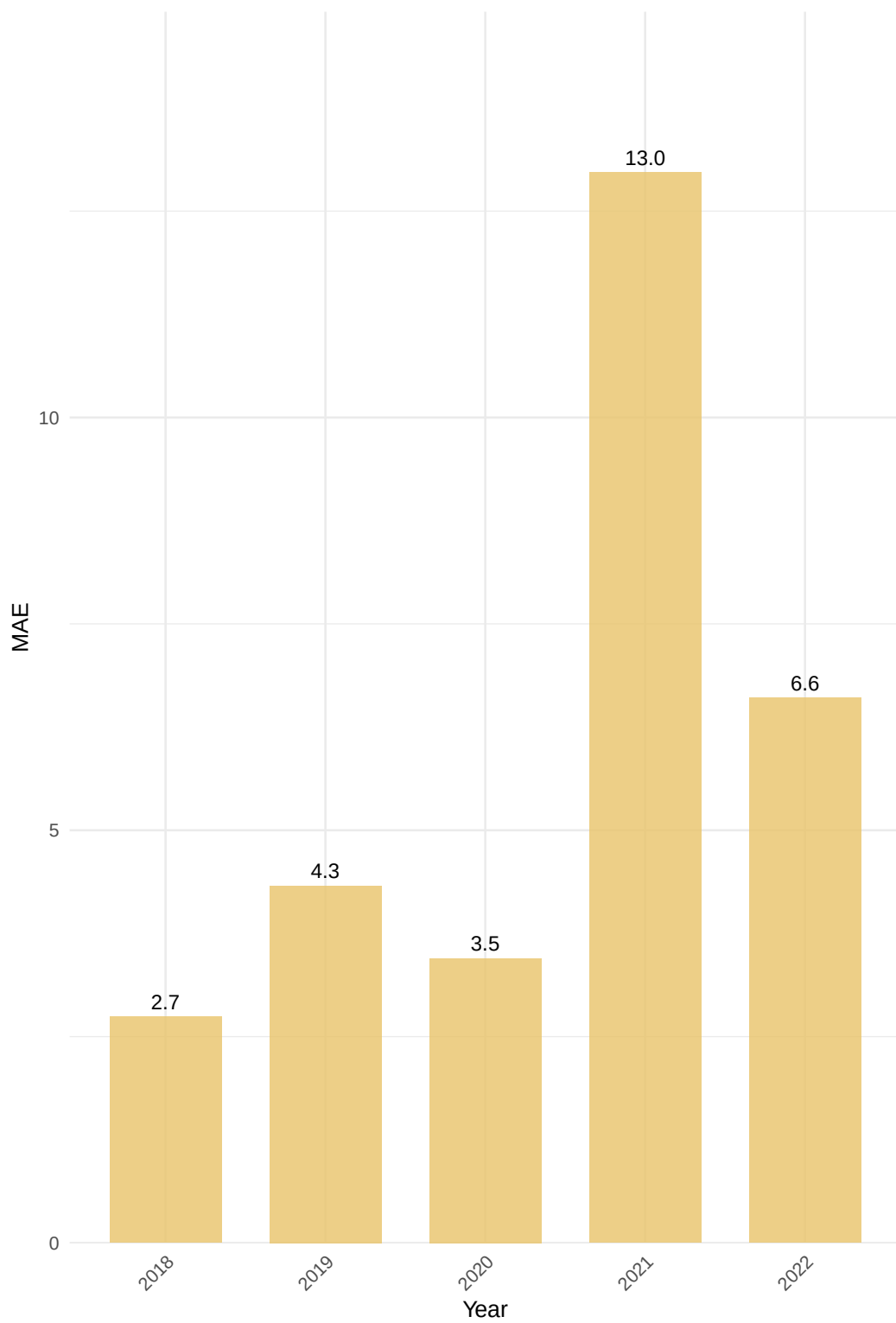


RMSE by Year (Original Scale)

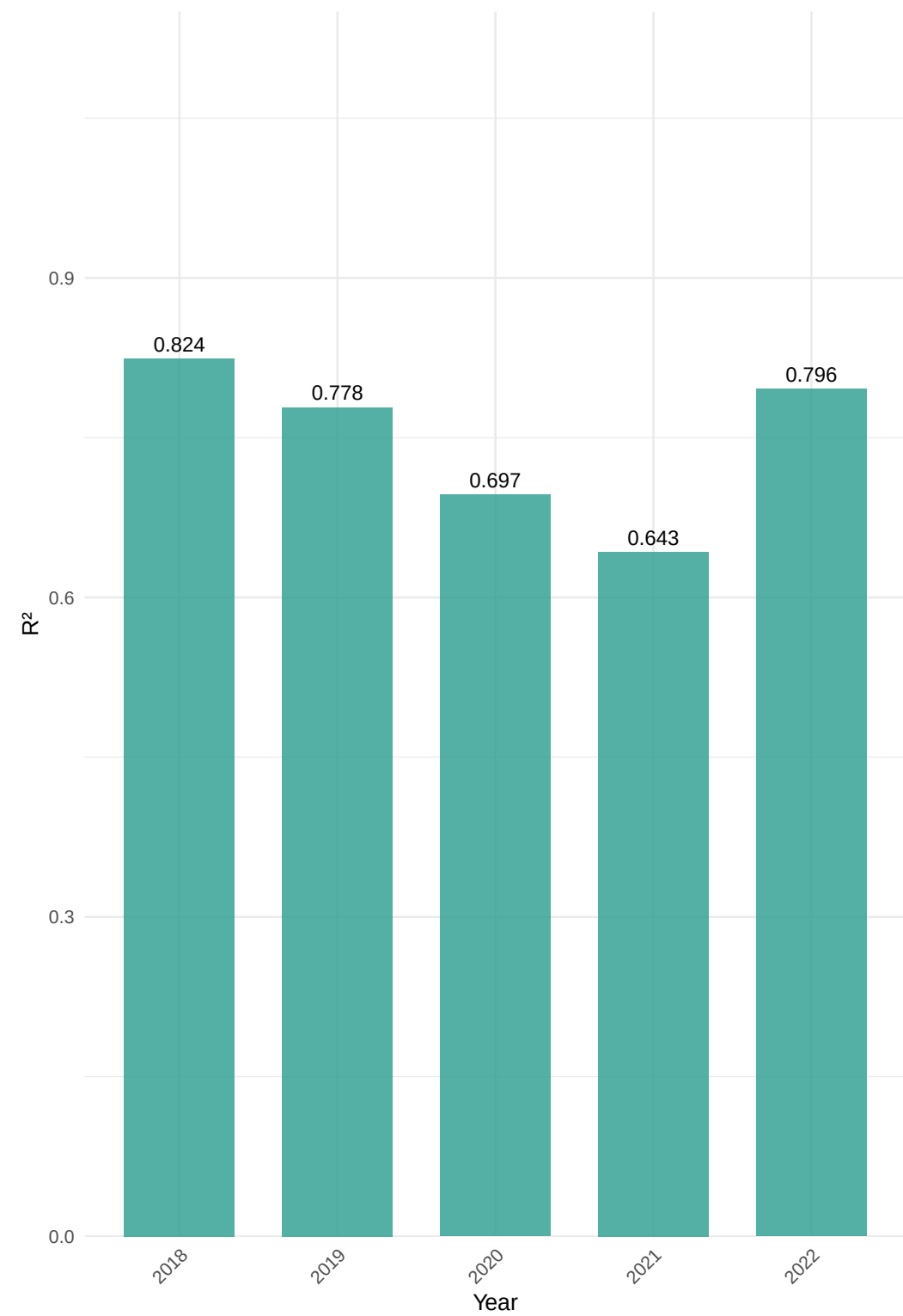


Performance Metrics by Year

MAE by Year

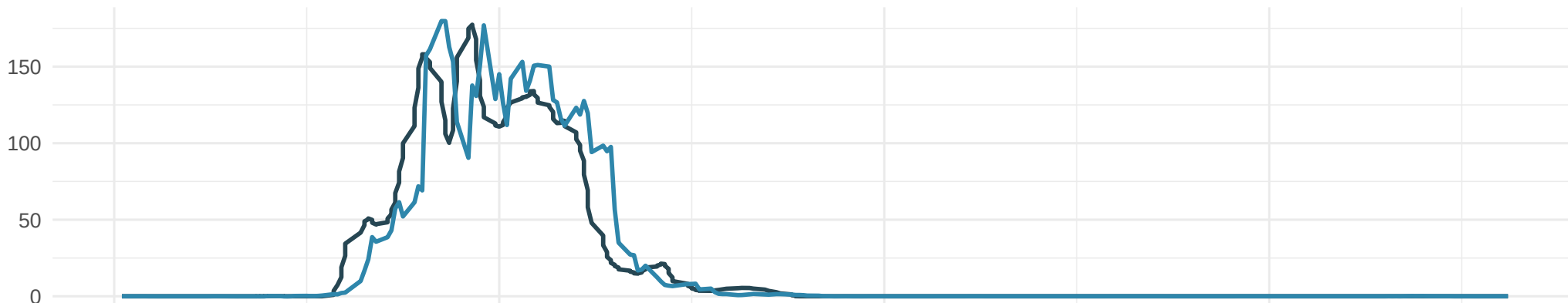


R² by Year

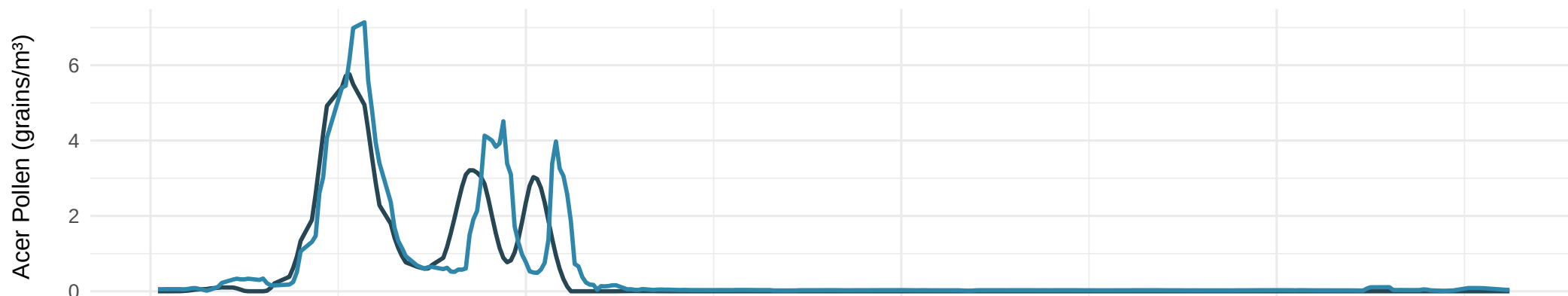


Sample Predictions – Year 2018 (Original Scale)

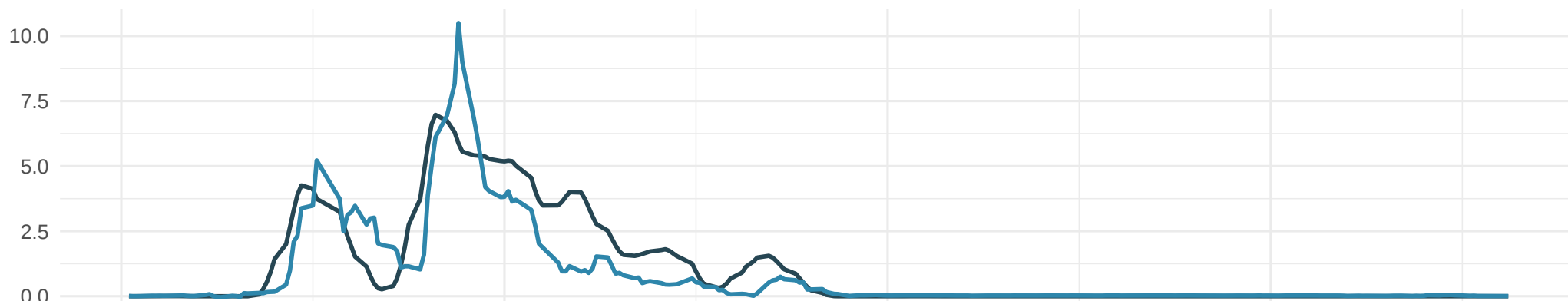
Lat: 31.51, Lon: -97.23



Lat: 33.94, Lon: -84.54



Lat: 35.55, Lon: -97.41



0

100

200

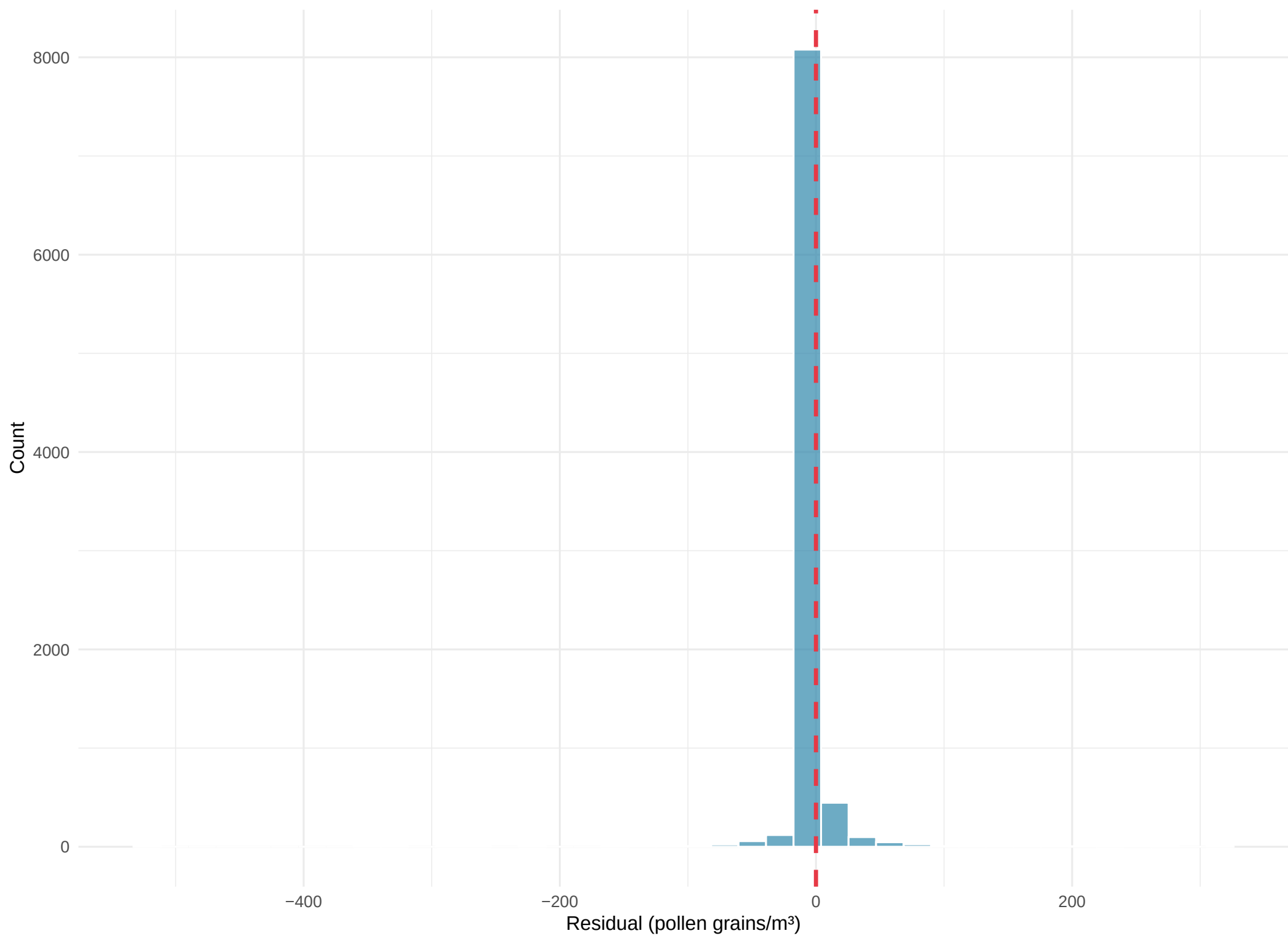
300

Day of Year

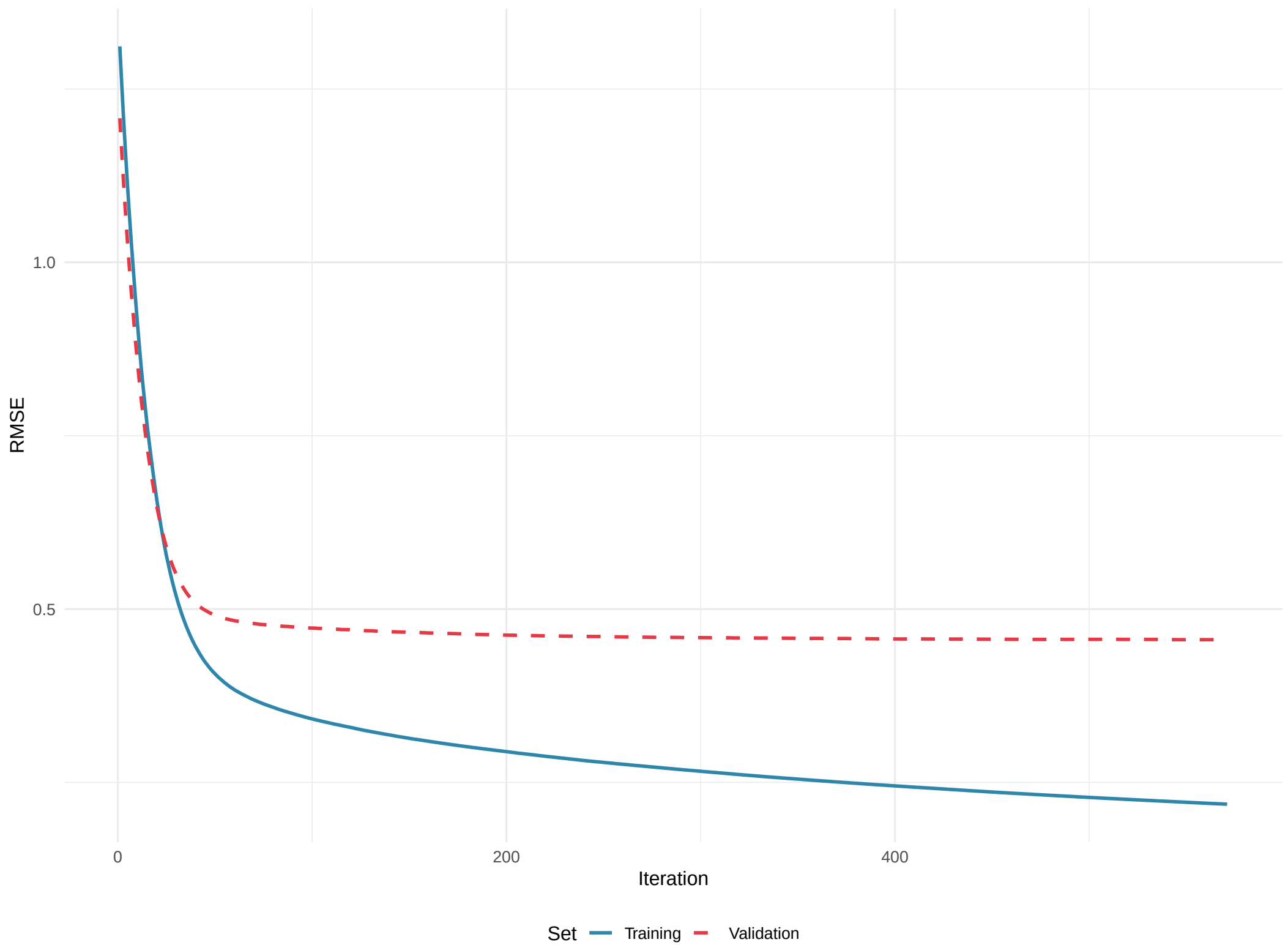
— Actual — Predicted

Residuals Distribution (Original Scale)

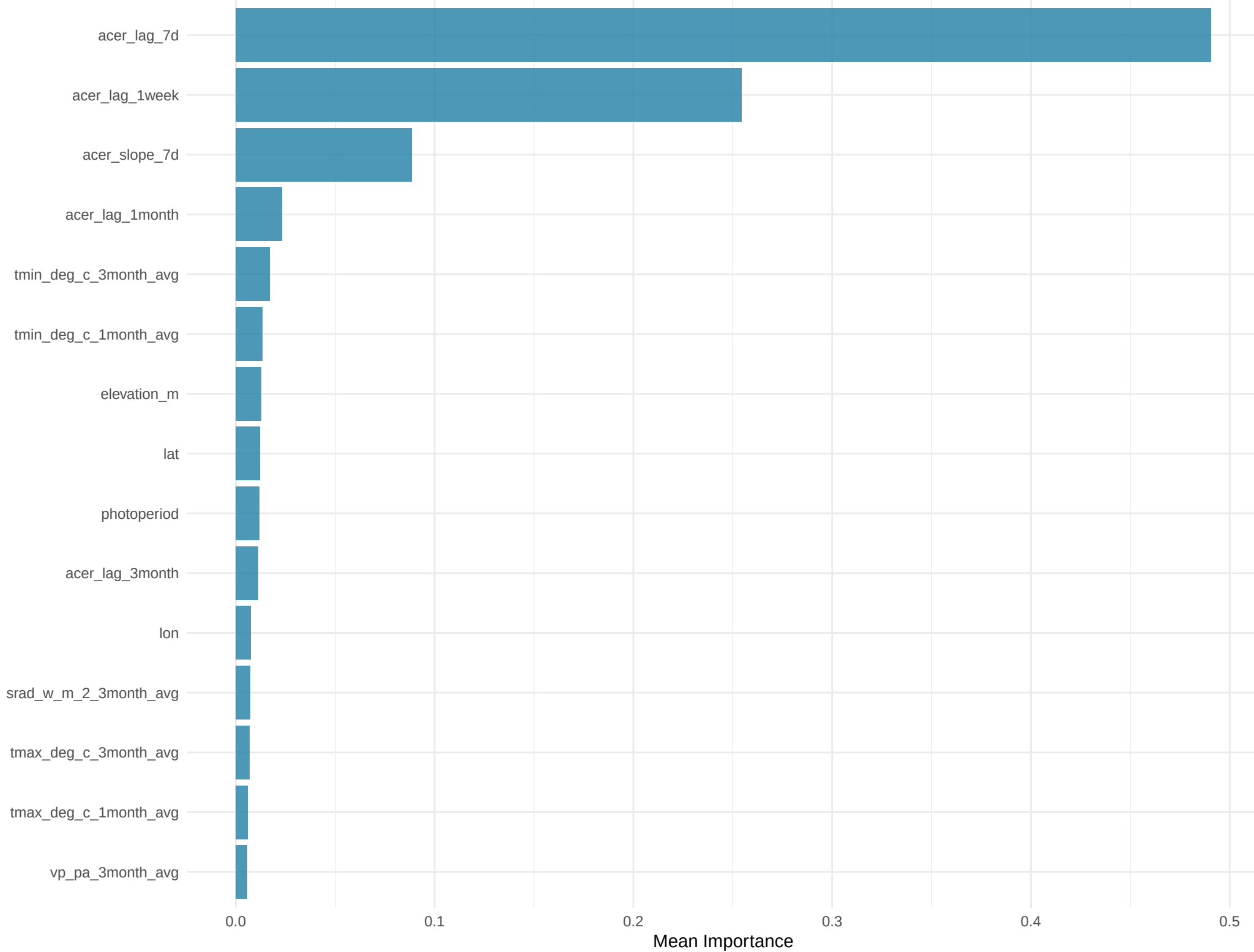
Mean: 0.34 | SD: 18.54 | Median: -0.03



Training vs Validation RMSE Curve



Top 15 Feature Importance (Avg Gain)



Summary Statistics

Test Set Performance (Original Scale)

RMSE: 18.54 pollen grains/m³

MAE: 4.24 pollen grains/m³

R²: 0.7631

Validation Performance

RMSE: 29.21 | MAE: 4.69 | R²: 0.7258

Duan Smearing Factor: 1.0257